Norm inflation in $\dot{H}^{1/4}(\mathbb{R}^3)$ for the defocusing quintic Gabitov–Turitsyn equation

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Abstract

We prove norm inflation in the homogeneous Sobolev space $\dot{H}^{1/4}(\mathbb{R}^3)$ for the defocusing quintic Gabitov–Turitsyn equation with averaging interval $[0,1]$. A family of Schwartz initial data has arbitrarily small initial norm, while its global smooth solutions attain arbitrarily large norms at arbitrarily small negative times. The proof combines a finite-interval virial identity with a lower bound for the averaged nonlinear forcing obtained from localized low-frequency mass. A rescaling keeps the variance bounded over a long time interval and forces growth of the fractional Sobolev norm.

1 Introduction

The Gabitov–Turitsyn equation is a nonlinear Schrödinger equation whose nonlinearity is averaged along the free Schrödinger flow. We consider the three-dimensional defocusing quintic equation

$$iu_t - \Delta u + Q_1(u) = 0, \qquad Q_1(f) = \int_0^1 T_{-\sigma}(|T_\sigma f|^4 T_\sigma f)\, d\sigma, \qquad T_\sigma = e^{i\sigma\Delta}. \tag{1.1}$$

The averaging interval is fixed throughout. For this sign and power, global well-posedness in the energy space with a finite second spatial moment follows from Kawakami and Murphy [1, Proposition 5.1]. This global control leaves open the behavior of weaker Sobolev norms for initial data whose energy is large.

Kowalski [3, Theorem 1.2] proves failure of C^5 regularity of the solution map below the mass-scaling index $s_m = 1$ for this model. Norm inflation asks for a stronger instability. The integrated-scaling index is $s_i = 1/2$; the range $0 < s < s_i$ is left open in the discussion of [3, Section 6.4]. We resolve the fixed exponent $s = 1/4$.

Theorem 1.1. *For every $\varepsilon > 0$, there are $u_0 \in \mathcal{S}(\mathbb{R}^3)$ and a global smooth solution u of (1.1), with $u(0) = u_0$, such that for some $t_\varepsilon \in (-\varepsilon, 0)$,*

$$\|u_0\|_{\dot{H}^{1/4}(\mathbb{R}^3)} < \varepsilon, \qquad \|u(t_\varepsilon)\|_{\dot{H}^{1/4}(\mathbb{R}^3)} > \varepsilon^{-1}.$$

The time in the theorem is negative and nonzero. The initial data are Gaussians with growing amplitude and shrinking width. Their H^1 norms need not be small, so the conclusion is compatible with global energy-space control.

Our argument uses the defocusing virial approach of Kowalski [3, Proposition 1.5 and Section 6.1], keeping the finite endpoint of the rescaled averaging interval. Growth of the kinetic energy alone does not give a lower bound for a fractional Sobolev norm: a small amount of mass at very high

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frequency can carry the kinetic energy. Instead, a bounded variance and a small fractional norm keep a fixed amount of mass in a fixed ball for a short free evolution. This yields a lower bound for a positive term in the variance identity. After rescaling, its time integral is bounded independently of the long observation interval. The resulting incompatibility forces growth in $\dot{H}^{1/4}$.

Section 2 proves the localization and virial estimates and justifies the regularity used in them. Section 3 gives the rescaling and the quantitative Gaussian construction. We make no claim here about the other positive exponents below $1/2$ or about inflation at positive times.

We use the unitary Fourier transform $\widehat{f}(\xi) = (2\pi)^{-3/2} \int_{\mathbb{R}^3} e^{-ix\cdot\xi} f(x)\, dx$ and

$$\|f\|_{\dot{H}^s}^2 = \int_{\mathbb{R}^3} |\xi|^{2s} |\widehat{f}(\xi)|^2 \, d\xi.$$

The spaces H^k are the inhomogeneous Sobolev spaces, $\|\cdot\|_p$ denotes the $L^p(\mathbb{R}^3)$ norm, and $\langle f, h\rangle = \int \bar{f} h$. All unspecified spatial integrals are over $\mathbb{R}^3$. Constants in the estimates may depend on a displayed derivative order or on fixed data, but the constants in the norm-inflation construction are independent of its parameter g.

2 Two estimates for the averaged equation

We first record the regularity needed for the virial calculation. Write $\Sigma = \{f \in H^1(\mathbb{R}^3) : xf \in L^2(\mathbb{R}^3)\}$.

Lemma 2.1 (Smooth Gaussian solutions)**.** *For $A, L > 0$ and $u_0(x) = A\exp(-|x|^2/(4L^2))$, equation (1.1) has a global smooth solution u. On every compact time interval I,*

$$u,\ xu,\ |x|^2 u \in C(I; H^k(\mathbb{R}^3)) \qquad (k = 0, 1, 2, \ldots).$$

Proof. Kawakami and Murphy [1, Proposition 5.1] prove global well-posedness in Σ, with a uniform H^1 bound, for the defocusing equation $iv_t + \Delta v = Q_1(v)$ when $d = 3$ and the nonlinearity is $|v|^4 v$. The change $u(t) = v(-t)$ gives (1.1) with the same averaging interval.

Here are persistence estimates for the smooth weighted data used below. Within this proof, D^k denotes the family of order-k derivatives. For each integer $k \geq 1$,

$$\|Q_1(f)\|_{H^k} \leq C_k \|f\|_{H^1}^4 \|f\|_{H^k}. \tag{2.1}$$

Indeed, the free Strichartz estimate [2, Theorem 1.2] for $(q, r) = (5, 30/11)$ and Sobolev embedding give

$$\|T_\sigma D^k f\|_{L_\sigma^5 L_x^{30/11}} \lesssim_k \|f\|_{H^k}, \qquad \|T_\sigma f\|_{L_\sigma^5 L_x^{30}} \lesssim \|\nabla f\|_2, \qquad 0 \leq \sigma \leq 1.$$

In a Leibniz term, let $k_1 + \cdots + k_5 = k$ be the derivative orders on its five factors. Integer Gagliardo–Nirenberg interpolation [4, p. 125] gives, with $h = T_\sigma f$,

$$\|D^{k_j} h\|_{p_j} \lesssim_k \|h\|_{30}^{1-k_j/k} \|D^k h\|_{30/11}^{k_j/k}, \qquad \frac{1}{p_j} = \frac{1 + 10k_j/k}{30}.$$

The cases $k_j = 0, k$ are identities. The spatial reciprocals sum to $1/2$; five L_σ^5 norms give an L_σ^1 product. Unitarity of $T_{-\sigma}$ proves the highest-derivative part of (2.1). For its L^2 part use one $L_x^{30/11}$ factor and four L_x^{30} factors. Summing the lower derivatives proves (2.1). The admissibility condition is $2/5 + 3/(30/11) = 3/2$.

For the weights, set $J_\sigma = x + 2i\sigma\nabla$ and $F(z) = z^3\bar{z}^2$. The identities

$$J_\sigma T_\sigma f = T_\sigma(xf), \qquad J_\sigma F(h) = 3h^2\bar{h}^2 J_\sigma h - 2h^3\bar{h}\,\overline{J_\sigma h}$$

and the H^k algebra property, $k \geq 2$, yield

$$\|xQ_1(f)\|_{H^k} \leq C_k \|f\|_{H^k}^4 \|xf\|_{H^k}. \tag{2.2}$$

Expanding $\sum_{j=1}^3 (x_j + 2i\sigma\partial_j)^2 F(T_\sigma f)$ also gives

$$\||x|^2 Q_1(f)\|_{H^k} \leq C_{k,R}(1 + \||x|^2 f\|_{H^k}) \quad \text{if } \|f\|_{H^{k+2}} + \|xf\|_{H^{k+1}} \leq R. \tag{2.3}$$

To see the dependence in this estimate, the free position commutator bounds $\||x|^2 T_\sigma f\|_{H^k}$ by a constant times $\||x|^2 f\|_{H^k} + \|xf\|_{H^{k+1}} + \|f\|_{H^{k+2}}$ for $0 \leq \sigma \leq 1$. The terms with $x\nabla F$ contain one weighted factor, and those with ΔF contain at most two further derivatives.

Local contraction can be run in the sum-norm space $H^{k+2} \cap \{xf \in H^{k+1},\ |x|^2 f \in H^k\}$; the corresponding polynomial difference estimates give local Lipschitz continuity. This solution lies in Σ and is identified with the global solution above. On a compact interval, (2.1) and the H^1 bound first control all higher Sobolev norms by Gronwall. The commutators

$$[\Delta, x] = 2\nabla, \qquad [\Delta, |x|^2] = 4x \cdot \nabla + 6$$

then let (2.2) control the first weight, followed by (2.3) for the second. These bounds give continuation in the same weighted space. Sobolev embedding and the equation give smoothness in space and time. □

The first estimate controls the positive term that will appear in the variance identity. A small fractional norm leaves most of the mass below a suitable frequency cutoff; its position norm stays bounded during a short free evolution.

Lemma 2.2 (Free mass localization)**.** *Fix $M, Y > 0$ and $0 < s < 1$. There is $c = c(M, Y, s) > 0$ such that, for $S \geq 1$ and $f \in H^1(\mathbb{R}^3)$ satisfying $\|f\|_2^2 = M$ and $\|xf\|_2 \leq Y$,*

$$W_S(f) := \int_0^S \sigma \|T_\sigma f\|_6^6 \, d\sigma \geq c(1 + \|f\|_{\dot{H}^s})^{-2/s}. \tag{2.4}$$

Proof. Choose $\chi \in C_c^\infty(\mathbb{R}^3)$ with $0 \leq \chi \leq 1$, $\chi = 1$ on $\{|\xi| \leq 1\}$, and $\chi = 0$ on $\{|\xi| \geq 2\}$. Let P_N be the Fourier multiplier with symbol $\chi(\xi/N)$, and set $f_N = P_N f$. Put $B = \|f\|_{\dot{H}^s}$ and

$$N = \max\{1, [16(1 + B)/\sqrt{M}]^{1/s}\}.$$

Plancherel gives

$$\|f - f_N\|_2 \leq N^{-s} B \leq \frac{\sqrt{M}}{16}, \qquad \|f_N\|_2 \geq \frac{15\sqrt{M}}{16}.$$

Differentiating the cutoff in frequency gives

$$\|xf_N\|_2 \leq Y + C_\chi \frac{\sqrt{M}}{N}, \qquad \|\nabla f_N\|_2 \leq 2N\sqrt{M}.$$

Since $xT_\sigma = T_\sigma(x - 2i\sigma\nabla)$, for $0 \leq \sigma \leq 1/N$,

$$\|xT_\sigma f_N\|_2 \leq Y + (C_\chi + 4)\sqrt{M} =: Y_*.$$

Fix $R \geq 1$ so that $Y_*/R \leq \sqrt{M}/8$. The mass outside the ball then satisfies $\|\mathbf{1}_{|x|>R} T_\sigma f_N\|_2 \leq \sqrt{M}/8$. Subtracting this tail and the high-frequency remainder gives

$$\|\mathbf{1}_{|x|\leq R} T_\sigma f\|_2 \geq (15/16 - 1/8 - 1/16)\sqrt{M} \geq \sqrt{M}/2.$$

Hölder on the ball $B_R = \{|x| \leq R\}$ yields

$$\|T_\sigma f\|_6^6 \geq \frac{(M/4)^3}{|B_R|^2} =: c_*.$$

The interval $[0, 1/N]$ lies in $[0, S]$, so $W_S(f) \geq c_*/(2N^2)$. Finally $N \leq C_{M,s}(1+B)^{1/s}$ proves (2.4). The cutoff is fixed once; all constants are independent of f and S. □

We next keep track of the finite upper endpoint in the averaged nonlinearity. For $S > 0$ write

$$Q_S(f) = \int_0^S T_{-\sigma}(|T_\sigma f|^4 T_\sigma f)\, d\sigma.$$

For a solution of

$$iw_\tau - \beta \Delta w + Q_S(w) = 0, \qquad \beta > 0, \tag{2.5}$$

let

$$V = \|xw\|_2^2, \qquad D = \operatorname{Im} \int_{\mathbb{R}^3} \bar{w}\, x \cdot \nabla w\, dx, \qquad G(\sigma, \tau) = \|T_\sigma w(\tau)\|_6^6,$$

$$P(\tau) = \int_0^S G(\sigma, \tau)\, d\sigma, \qquad W(\tau) = W_S(w(\tau)).$$

Here V is the second spatial moment and D is its dilation moment. The mass and positive energy are

$$M = \|w\|_2^2, \qquad E = \frac{\beta}{2}\|\nabla w\|_2^2 + \frac{1}{6}P. \tag{2.6}$$

Lemma 2.3 (Finite-interval virial identities)**.** *Suppose w solves (2.5) on a time interval I and $w, xw, |x|^2 w \in C(I; H^k)$ for every integer $k \geq 0$. Then M, E are constant and*

$$V' = -4\beta D + \frac{16}{3}W + \frac{4}{3}S^2 G(S, \tau), \tag{2.7}$$

$$D' = -2\beta\|\nabla w\|_2^2 - \frac{4}{3}P - \frac{2}{3}SG(S, \tau) = -4E - \frac{2}{3}P - \frac{2}{3}SG(S, \tau). \tag{2.8}$$

Proof. Unitarity gives $\langle w, Q_S(w)\rangle = P \in \mathbb{R}$, which proves mass conservation. In the real L^2 pairing, the derivative of E is $-\beta\Delta w + Q_S(w)$. Equation (2.5) reads $w_\tau = i(-\beta\Delta w + Q_S(w))$, so $E' = 0$.

For the variance calculation set $h = T_\sigma w$. The position commutator gives

$$T_\sigma |x|^2 w = (|x|^2 + 4i\sigma x \cdot \nabla + 6i\sigma - 4\sigma^2\Delta)h.$$

The nonlinear part of V' is -2 times the imaginary part of the pairing of this expression with $|h|^4 h$. Integration by parts gives

$$\operatorname{Re} \int x \cdot \nabla \bar{h}\, |h|^4 h = -G/2, \qquad \partial_\sigma G = 6 \operatorname{Im} \int \Delta \bar{h}\, |h|^4 h.$$

Its integrand is therefore $8\sigma G + (4/3)\sigma^2 \partial_\sigma G$. After integration in σ this becomes $(16/3)W + (4/3)S^2 G(S, \tau)$. The linear contribution is $-4\beta D$, proving (2.7).

For (2.8), use $w_\lambda(x) = \lambda^{3/2} w(\lambda x)$. Then

$$P(w_\lambda) = \lambda^4 \int_0^{\lambda^2 S} G(\sigma, \tau)\, d\sigma, \qquad \frac{d}{d\lambda} P(w_\lambda)\Big|_{\lambda=1} = 4P + 2SG(S,\tau).$$

The generator $\mathcal{A} = x \cdot \nabla + 3/2$ is skew-adjoint, and $D = \operatorname{Im}\langle w, \mathcal{A} w\rangle$. Consequently

$$D' = 2 \operatorname{Im}\langle w_\tau, \mathcal{A} w\rangle = -2 \frac{d}{d\lambda} E(w_\lambda)\Big|_{\lambda=1}.$$

Differentiating the kinetic term and the preceding formula for P gives (2.8). The assumed weighted regularity justifies these pairings, dilations and finite-interval integrations by parts. □

3 Proof of norm inflation

Let $\phi(y) = e^{-|y|^2/4}$, and, for $g \geq 2$, choose

$$L = g^{-9/4}, \qquad A = g^{11/4}, \qquad u_{g,0}(x) = A\phi(x/L). \tag{3.1}$$

Let u_g be the smooth solution given by Lemma 2.1. Define

$$w_g(\tau, y) = A^{-1} u_g(\tau/(A^4 L^2), Ly).$$

For a scaled function, $T_\sigma[Af(\cdot/L)](x) = A(T_{\sigma/L^2} f)(x/L)$. Changing the integration variable gives

$$Q_1[Af(\cdot/L)](x) = A^5 L^2 (Q_{L^{-2}} f)(x/L).$$

Thus w_g satisfies (2.5) exactly, with

$$\beta = (AL)^{-4} = g^{-2}, \qquad S = L^{-2} = g^{9/2}, \qquad \tau = g^{13/2} t, \qquad w_g(0) = \phi. \tag{3.2}$$

The averaging interval remains finite. Fourier scaling also gives

$$\|u_g(t)\|_{\dot{H}^{1/4}} = AL^{5/4} \|w_g(\tau)\|_{\dot{H}^{1/4}} = g^{-1/16} \|w_g(\tau)\|_{\dot{H}^{1/4}}. \tag{3.3}$$

We use the variance and energy of w_g on $[-g, 0]$. Suppress its index, and set $M_0 = \|\phi\|_2^2 > 0$ and $V_0 = \|y\phi\|_2^2$. The free Gaussian satisfies

$$T_\sigma \phi = (1 + i\sigma)^{-3/2} \exp\left(-\frac{|y|^2}{4(1+i\sigma)}\right), \qquad \|T_\sigma \phi\|_6^6 = (2\pi/3)^{3/2} (1+\sigma^2)^{-3}.$$

Since $\int_0^\infty (1+\sigma^2)^{-3} d\sigma = 3\pi/16$, the initial energy, and hence the conserved energy, is bounded independently of g by

$$E \leq E_* := \tfrac{1}{2} \|\nabla \phi\|_2^2 + \frac{\pi}{32} (2\pi/3)^{3/2}. \tag{3.4}$$

The Gaussian is real, so $D(0) = 0$. Since $0 \leq P \leq 6E$, (2.8) gives $D' \geq -8E - (2/3) SG(S,\tau)$. For $\tau \leq r \leq 0$,

$$D(r) \leq -8Er + \frac{2}{3} S \int_r^0 G(S, q)\, dq.$$

Integrating (2.7) backwards gives

$$V(\tau) = V_0 + 4\beta \int_\tau^0 D(r)\, dr - \int_\tau^0 \left(\frac{16}{3} W(r) + \frac{4}{3} S^2 G(S, r)\right) dr.$$

The double integral of $G(S,q)$ equals $\int_\tau^0 (q-\tau)G(S,q)dq$. Combining these formulas yields

$$V(\tau) \le V_0 + 16\beta E\tau^2 - \frac{16}{3}\int_\tau^0 W(r)\,dr + \frac{4}{3}S\int_\tau^0 [2\beta(r-\tau) - S]G(S,r)\,dr. \tag{3.5}$$

For $-g \le \tau \le 0$, we have $2\beta g \le S$ and $\beta\tau^2 \le 1$. Both integral terms on the right are nonpositive. Therefore

$$V(\tau) \le V_* := V_0 + 16E_*, \qquad -g \le \tau \le 0. \tag{3.6}$$

Energy and Cauchy–Schwarz now give

$$\|\nabla w(\tau)\|_2 \le \sqrt{2E_*}\,g, \qquad |D(\tau)| \le C_D g, \qquad C_D = \sqrt{2E_*V_*}.$$

In the integral of (2.7) from $-g$ to zero the endpoint term is nonnegative. Using $V(-g) \ge 0$, we obtain the uniform bound

$$\frac{16}{3}\int_{-g}^0 W(r)\,dr \le V_0 + 4\beta\int_{-g}^0 D(r)\,dr \le V_0 + 4C_D =: \frac{16}{3}C_W. \tag{3.7}$$

Let $B_g = \max_{-g\le\tau\le 0}\|w_g(\tau)\|_{\dot H^{1/4}}$. This maximum is finite and attained by H^1 continuity. Lemma 2.2 applies at every time with the same mass M_0, position bound $\sqrt{V_*}$, and $s = 1/4$. Its constant is independent of g, and it gives

$$cg(1+B_g)^{-8} \le \int_{-g}^0 W(r)\,dr \le C_W.$$

Thus, for all sufficiently large g,

$$B_g \ge (cg/C_W)^{1/8} - 1 \ge c_1 g^{1/8}, \tag{3.8}$$

where $c_1 > 0$ is fixed. A maximizing time τ_g is different from zero once this lower bound exceeds $\|\phi\|_{\dot H^{1/4}}$. Set $t_g = \tau_g g^{-13/2} < 0$. Equations (3.3) and (3.8) then give

$$\|u_{g,0}\|_{\dot H^{1/4}} = g^{-1/16}\|\phi\|_{\dot H^{1/4}}, \qquad \|u_g(t_g)\|_{\dot H^{1/4}} \ge c_1 g^{1/16}, \qquad 0 < |t_g| \le g^{-11/2}.$$

Choosing g large enough makes the first norm less than ε, the second larger than ε^{-1}, and the nonzero time less than ε in absolute value. This proves Theorem 1.1.

The construction can be recorded without the rescaled quantities.

Corollary 3.1 (Gaussian construction)**.** *There are constants $c > 0$ and $g_0 \ge 2$ such that, for every real $g \ge g_0$, the global smooth solution of (1.1) with*

$$u_{g,0}(x) = g^{11/4}\exp(-g^{9/2}|x|^2/4), \qquad x \in \mathbb{R}^3,$$

has a time $-g^{-11/2} \le t_g < 0$ satisfying

$$\|u_{g,0}\|_{\dot H^{1/4}} = C_\phi g^{-1/16}, \qquad \|u_g(t_g)\|_{\dot H^{1/4}} \ge c g^{1/16}.$$

Here $C_\phi = \|e^{-|\cdot|^2/4}\|_{\dot H^{1/4}}$, $u_g(0) = u_{g,0}$, and c, g_0 depend only on the fixed Gaussian and the fixed cutoff in Lemma 2.2.

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References

[1] Jumpei Kawakami and Jason Murphy, *Small and large data scattering for the dispersion-managed NLS*, arXiv:2407.11151v1 (2024), https://arxiv.org/abs/2407.11151v1.

[2] Markus Keel and Terence Tao, *Endpoint Strichartz estimates*, American Journal of Mathematics **120** (1998), 955–980.

[3] Matthew Kowalski, *On ill-posedness for the Gabitov–Turitsyn equation*, arXiv:2510.11887v1 (2025), https://arxiv.org/abs/2510.11887v1.

[4] L. Nirenberg, *On elliptic partial differential equations*, Annali della Scuola Normale Superiore di Pisa, Classe di Scienze, Series 3, **13** (1959), no. 2, 115–162.